

CANDIDATE
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9618/22

May/June 2023

2 hours

You must answer on the question paper.

You will need: Insert (enclosed)

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.
- The insert contains all the resources referred to in the questions.

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[Turn over

Refer to the **insert** for the list of pseudocode functions and operators.

- 1 A program calculates the postal cost based on the weight of the item and its destination. Calculations occur at various points in the program and these result in the choice of several possible postal costs. The programmer has built these postal costs into the program.

For example, the postal cost of \$3.75 is used in the following lines of pseudocode:

```
IF Weight < 250 AND ValidAddress = TRUE THEN
    ItemPostalCost ← 3.75      // set postal cost for item to $3.75
    ItemStatus ← "Valid"      // item can be sent
ENDIF
```

- (a) (i) Identify a more appropriate way of representing the postal costs.

..... [1]

- (ii) Describe the advantages of your answer to **part (a)(i)** with reference to this program.

.....

 [3]

- (b) The lines of pseudocode contain features that make them easier to understand.

State **three** of these features.

1
 2
 3 [3]

- (c) Give the **appropriate** data types for the following variables:

ValidAddress
 ItemPostalCost
 ItemStatus [3]

2 A program stores a user's date of birth using a variable `MyDOB` of type `DATE`.

- (a) Write a pseudocode statement, using a function from the **insert**, to assign the value corresponding to 17/11/2007 to `MyDOB`.

..... [1]

- (b) `MyDOB` has been assigned a valid value representing the user's date of birth.

Write a pseudocode statement to calculate the number of months from the month of the user's birth until the end of the year and to assign this to the variable `NumMonths`.

For example, if `MyDOB` contains a value representing 02/07/2008, the value 5 would be assigned to `NumMonths`.

..... [2]

- (c) The program will output the day of the week corresponding to `MyDOB`.

For example, given the date 22/06/2023, the program will output "Thursday".

An algorithm is required. An array will be used to store the names of the days of the week.

Define the array **and** describe the algorithm in **four** steps.

Do **not** use pseudocode statements in your answer.

Array definition

.....

Step 1

.....

.....

Step 2

.....

.....

Step 3

.....

.....

Step 4

.....

.....

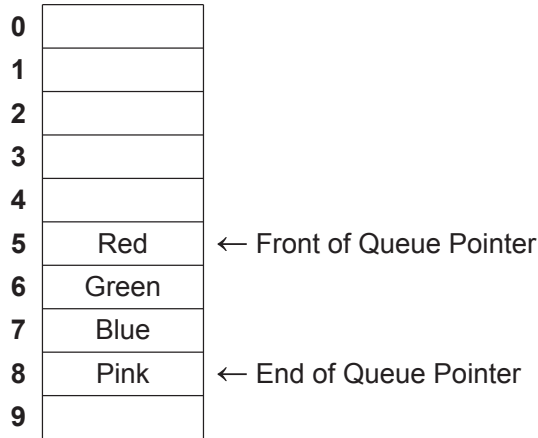
[6]

3 A program stores data in a text file. When data is read from the file, it is placed in a queue.

- (a) The diagram below represents an Abstract Data Type (ADT) implementation of the queue. Each data item is stored in a separate location in the data structure. During initial design, the queue is limited to holding a maximum of 10 data items.

The operation of this queue may be summarised as follows:

- The Front of Queue Pointer points to the next data item to be removed.
- The End of Queue Pointer points to the last data item added.
- The queue is circular so that locations can be reused.



- (i) Describe how the data items Orange and Yellow are added to the queue shown in the diagram.

.....

.....

.....

.....

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..... [4]

- (ii) The following diagram shows the state of the queue after several operations have been performed. All queue locations have been used at least once.

0	D4	
1	D3	← End of Queue Pointer
2	D27	
3	D8	
4	D33	
5	D17	← Front of Queue Pointer
6	D2	
7	D1	
8	D45	
9	D60	

State the number of data items in the queue.

..... [1]

- (b)** The design of the queue is completed and the number of locations is increased.

A function `AddToQueue()` has been written. It takes a string as a parameter and adds this to the queue. The function will return `TRUE` if the string was added successfully.

A procedure `FileToQueue()` will add each line from the file to the queue. This procedure will end when all lines have been added or when the queue is full.

Describe the algorithm for procedure `FileToQueue()`.

Do **not** use pseudocode in your answer.

..... [5]

4 A function `GetNum()` will:

1. take two parameters: a string and a character
2. count the number of times that the character occurs in the string
3. return the count.

Any comparison between characters needs to be case sensitive. For example, character 'a' and character 'A' are not identical.

Write pseudocode for function `GetNum()`.

[6]

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- 5 A programmer has produced the following pseudocode to output the square root of the numbers from 1 to 10.

Line numbers are for reference only.

```

10  DECLARE Num : REAL
11  Num ← 1.0
...
40  REPEAT
41      CALL DisplaySqrt(Num)
42      Num ← Num + 1.0
43  UNTIL Num > 10
...
50  PROCEDURE DisplaySqrt(BYREF ThisNum : REAL)
51      OUTPUT ThisNum
52      ThisNum ← SQRT(ThisNum)      // SQRT returns the square root
53      OUTPUT " has a square root of ", ThisNum
54  ENDPROCEDURE

```

The pseudocode is correctly converted into program code.

Function `SQRT()` is a library function and contains no errors.

The program code compiles without errors, but the program gives unexpected results. These are caused by a design error.

- (a) Explain why the program gives unexpected results.

.....

.....

.....

.....

.....

..... [3]

- (b) Explain why the compiler does **not** identify this error.

.....

..... [1]

- (c) Describe how a typical Integrated Development Environment (IDE) could be used to identify this error.

.....

.....

.....

.....

.....

..... [3]

- (d) The pseudocode is converted into program code as part of a larger program.

During compilation, a complex statement generates an error.

The programmer does not want to delete the complex statement but wants to change the statement so that it is ignored by the compiler.

State how this may be achieved.

.....

..... [1]

- 6 A procedure `Square()` will take an integer value in the range 1 to 9 as a parameter and output a number square.

The boundary of a number square is made up of the character representing the parameter value. The inside of the number square is made up of the asterisk character (*).

Parameter value	1	2	3	4	...	9
Output	1	22 22	333 3*3 333	4444 4**4 4**4 4444	...	999999999 9*****9 9*****9 9*****9 9*****9 9*****9 9*****9 9*****9 999999999

The pseudocode `OUTPUT` command starts each output on a new line. For example, the following three `OUTPUT` statements would result in the outputs as shown:

```
OUTPUT "Hello"
OUTPUT "ginger"
OUTPUT "cat"
```

Resulting output:

```
Hello
ginger
cat
```

Write pseudocode for procedure `Square()`.

Parameter validation is not required.

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..... [6]

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- 7 A computer system for a shop stores information about each customer. The items of information include name and address (both postal and email) together with payment details and order history. The system also stores the product categories they are interested in and how they would like to be contacted.

- (a) The shop wants to add a program module that will generate emails to be sent to customers who may be interested in receiving details of new products.

- (i) State **three** items of information that the new module would need. Justify your choice in each case.

Information

Justification

.....

Information

Justification

.....

Information

Justification

.....

[3]

- (ii) Identify **two** items of customer information that would **not** be required by the new module. Justify your choice in each case.

Information

Justification

.....

Information

Justification

.....

[2]

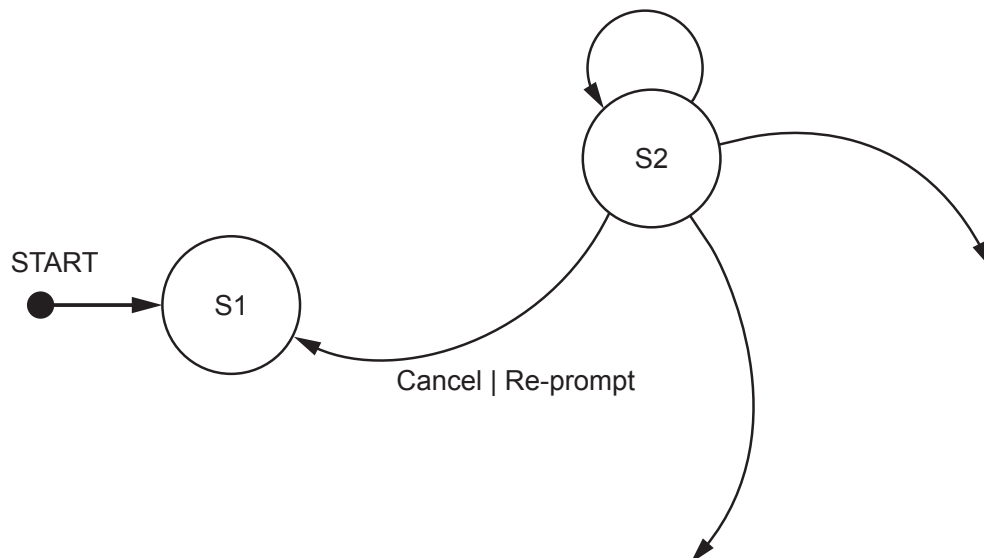
- (b) The program includes a module to validate a Personal Identification Number (PIN). This is used when customers pay for goods using a bank card.

A state-transition diagram has been produced for this module.

The table show the inputs, outputs and states for this part of the program:

Current state	Input	Output	Next state
S1	Input PIN		S2
S2	Re-input PIN	Display error	S2
S2	Cancel	Re-prompt	S1
S2	Valid PIN	Enable payment	S4
S2	Too many tries	Block Account	S3

Complete the state-transition diagram to represent the information given in the table.



- 8 A computer shop assembles computers using items bought from several suppliers. A text file `Stock.txt` contains information about each item.

Information for each item is stored as a single line in the `Stock.txt` file in the format:

`<ItemNum><SupplierCode><Description>`

Valid item information is as follows:

	Format	Comment
ItemNum	4 numeric characters	unique number for each item in the range "0001" to "5999" inclusive
SupplierCode	3 alphabetic characters	to identify the supplier of the item
Description	a string	a minimum of 12 characters

The file is organised in ascending order of `ItemNum` and does **not** contain all possible values in the range.

A programmer has started to define program modules as follows:

Module	Description
<code>OnlyAlpha()</code> (already written)	- called with a parameter of type string - returns <code>TRUE</code> if the string contains only alphabetic characters, otherwise returns <code>FALSE</code>
<code>CheckInfo()</code>	- called with a parameter of type string representing a line of item information - checks to see whether the item information in the string is valid - returns <code>TRUE</code> if the item information is valid, otherwise returns <code>FALSE</code>

- (a) Write pseudocode for module `CheckInfo()`.

Module `OnlyAlpha()` should be used as part of your solution.

[illegible]

Module	Description
AddItem()	- called with a parameter of type string representing valid information for a new item that is not currently in the <code>Stock.txt</code> file - creates a new file <code>NewStock.txt</code> from the contents of the file <code>Stock.txt</code> and adds the new item information at the appropriate place in the <code>NewStock.txt</code> file

Write pseudocode for module `AddItem()`.

[illegible]

Identify a method that can be used to continue testing the main program before the errors in these modules have been corrected **and** describe how this would work.

Method

Description

.....

[3]

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